

$$1 \text{ (a)} \quad (B - B^3 + B^5)X_t = BX_t - B^3X_t + B^5X_t$$

$$= X_{t-1} - X_{t-3} + X_{t-5}$$

B^k - shifts X_t
k steps back

$$(c) \quad \nabla(1 + B - B^2)X_t = \nabla[X_t + X_{t-1} - X_{t-2}]$$

$$= \nabla X_t + \nabla X_{t-1} - \nabla X_{t-2}$$

$$= (X_t - X_{t-1})$$

$$+ (X_{t-1} - X_{t-2})$$

$$- (X_{t-2} - X_{t-3})$$

$$= X_t - 2X_{t-2} + X_{t-3}$$

$$2 \text{ (a)} \quad \left(1 - \frac{1}{3}B\right)X_t = Z_t$$

$$\Rightarrow X_t = \frac{1}{\left(1 - \frac{1}{3}B\right)} Z_t$$

$$= \left(1 - \frac{B}{3}\right)^{-1} Z_t$$

Let

$$f(B) = \left(1 - \frac{B}{3}\right)^{-1}$$

$$f'(B) = \left(\frac{1}{3}\right)\left(1 - \frac{B}{3}\right)^{-2}$$

MA(∞) form

$$X_t = \sum_{j=0}^{\infty} \beta_j Z_{t-j}$$

Taylor series exp. of $\left(1 - \frac{B}{3}\right)^{-1}$

$$f''(B) = 2 \left(\frac{1}{3}\right)^2 \left(1 - \frac{B}{3}\right)^{-3}$$

Generally if $f^{(k)}(B)$ is the k^{th} derivative of $f(B)$ then

$$f^{(k)}(B) = k! \left(\frac{1}{3}\right)^k \left(1 - \frac{B}{3}\right)^{-(k+1)}$$

Taylor series expansion of $f(B)$ about 0 is

$$f(B) = f(0) + \frac{f'(0)}{1!} B + \frac{f''(0)}{2!} B^2 + \dots$$

$$\therefore f(B) = \sum_{j=0}^{\infty} \left(\frac{1}{3}\right)^j B^j$$

$$\Rightarrow X_t = \sum_{j=0}^{\infty} \left(\frac{1}{3}\right)^j B^j Z_t$$

$$= \sum_{j=0}^{\infty} \left(\frac{1}{3}\right)^j Z_{t-j}.$$

3. $\{X_t\}$ - mean zero $\mathbb{E}(X_t) = 0 \quad \forall t$
- autocovariance f'n $\gamma_X(h)$
- stationary.

$$\begin{aligned} (a) \quad \mathbb{E}(Y_t) &= \mathbb{E}(X_t - X_{t-1}) \\ &= \mathbb{E}(X_t) - \mathbb{E}(X_{t-1}) \\ &= 0. \end{aligned}$$

(b) For $h = 0, \pm 1, \pm 2, \dots$

$$\begin{aligned} \gamma_Y(h) &= \text{Cov}(Y_t, Y_{t+h}) \\ &= \mathbb{E}(Y_t Y_{t+h}) - \mathbb{E}(X_t) \mathbb{E}(Y_{t+h}) \\ &= \mathbb{E}(Y_t Y_{t+h}) \\ &= \mathbb{E}[(X_t - X_{t-1})(X_{t+h} - X_{t+h-1})] \\ &= \mathbb{E}(X_t X_{t+h}) - \mathbb{E}(X_{t+h} X_{t-1}) \\ &\quad - \mathbb{E}(X_t X_{t+h-1}) + \mathbb{E}(X_{t-1} X_{t+h-1}) \\ &= \gamma_X(h) - \gamma_X(h+1) - \gamma_X(h-1) \\ &\quad + \gamma_X(h) \\ &= 2\gamma_X(h) - \gamma_X(h+1) - \gamma_X(h-1). \end{aligned}$$

4(a)

Given $X_t = Z_t + C(\bar{Z}_{t-1} + \bar{Z}_{t-2} + \dots)$

This is MA(∞) form where

$$X_t = \sum_{j=0}^{\infty} \psi_j \bar{Z}_{t-j}$$

MA(∞)
form

where $\psi_0 = 1$; $\psi_j = C$ for $j \geq 1$.

We know that a necessary condition for stationarity is that

$$\sum_{j=0}^{\infty} \psi_j^2 < \infty \quad (\text{i.e. finite sum})$$

Here
$$\sum_{j=0}^{\infty} \psi_j^2 = 1 + \sum_{j=1}^{\infty} C^2$$

$$= 1 + C^2 + C^2 + C^2 + \dots$$

is not finite unless $C = 0$.

$\therefore \{X_t\}$ is not weakly stationary unless $C = 0$.

$$\begin{aligned}
 (b) \quad Y_t &= X_t - X_{t-1} \\
 &= \bar{z}_t + C(\bar{z}_{t-1} + \bar{z}_{t-2} + \dots) \\
 &\quad - [\bar{z}_{t-1} + C(\bar{z}_{t-2} + \bar{z}_{t-3} + \dots)] \\
 &= \bar{z}_t + (C-1)\bar{z}_{t-1} \quad \leftarrow \text{MA(1) process} \\
 &= \bar{z}_t + \underbrace{\theta_1}_{C-1} \bar{z}_{t-1}
 \end{aligned}$$

Autocovariance f'n:

$$\begin{aligned}
 \gamma(0) &= \text{Var}(Y_t) \\
 &= \text{Var}[\bar{z}_t + \theta_1 \bar{z}_{t-1}] \\
 &= \sigma_{\bar{z}}^2 (1 + \theta_1^2) \quad \text{「 } \bar{z}_t \text{ and } \bar{z}_{t-1} \text{ are independent,} \\
 &= \sigma_{\bar{z}}^2 [1 + (C-1)^2]
 \end{aligned}$$

At lag 1

$$\begin{aligned}
 \gamma(1) &= \text{Cov}(Y_t, Y_{t-1}) \\
 &= \mathbb{E}(Y_t Y_{t-1}) \quad \text{「 } Y_t \text{ has mean zero,} \\
 &= \mathbb{E}[(\bar{z}_t + \theta_1 \bar{z}_{t-1})(\bar{z}_{t-1} + \theta_1 \bar{z}_{t-2})] \\
 &= \mathbb{E}[\cancel{\bar{z}_t \bar{z}_{t-1}}] + \theta_1 \mathbb{E}(\bar{z}_{t-1}^2) \\
 &\quad + \theta_1 \mathbb{E}(\cancel{\bar{z}_t \bar{z}_{t-2}}) + \theta_1^2 \mathbb{E}(\cancel{\bar{z}_{t-1} \bar{z}_{t-2}}) \\
 &= \theta_1 \mathbb{E}(\bar{z}_{t-1}^2) \\
 &= \sigma_{\bar{z}}^2 \theta_1 \quad \text{「 because } \mathbb{E}(\bar{z}_{t-1}^2) = \text{Var}(\bar{z}_t) \\
 &= \sigma_{\bar{z}}^2 (C-1)
 \end{aligned}$$

$\{Y_t\}$ is an MA(1) process

$\therefore \{Y_t\}$ is (weakly) stationary

$$\therefore \gamma(1) = \gamma(-1)$$

For an MA(1) process $\gamma(h) = 0$ for $|h| \geq 2$. (Lecture 3)

$$\therefore \gamma(h) = \begin{cases} \sigma_z^2 (1 + (c-1)^2) & h=0 \\ \sigma_z^2 (c-1) & h=\pm 1 \\ 0 & |h| \geq 2 \end{cases}$$

$\therefore$ The autocorrelation fun is $\rho(h) = \frac{\gamma(h)}{\gamma(0)}$.

$$\rho(h) = \begin{cases} 1 & h=0; \\ \frac{c-1}{1 + (c-1)^2} & h=\pm 1; \\ 0 & |h| \geq 2. \end{cases}$$

$$\begin{aligned}
 5. (a) \quad \mathbb{E}(X_t \bar{z}_{t-h}) &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \bar{z}_{t-j}\right) \bar{z}_{t-h}\right] \\
 &= \sum_{j=0}^{\infty} \psi_j \mathbb{E}(\bar{z}_{t-j} \bar{z}_{t-h})
 \end{aligned}$$

We know that $\mathbb{E}(\bar{z}_{t-j} \bar{z}_{t-h}) = 0$ where $j \neq h$ because $\{\bar{z}_t\}$ is a white noise process.

$$\begin{aligned}
 \therefore \mathbb{E}(X_t \bar{z}_{t-h}) &= \psi_h \mathbb{E}(\bar{z}_{t-h}^2) \\
 &= \sigma_z^2 \psi_h.
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \mathbb{E}(X_t X_{t-h}) &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \bar{z}_{t-j}\right) \left(\sum_{k=0}^{\infty} \psi_k \bar{z}_{t-h-k}\right)\right] \\
 &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \psi_j \psi_k \mathbb{E}(\bar{z}_{t-j} \bar{z}_{t-h-k})
 \end{aligned}$$

$$\begin{aligned}
 \mathbb{E}(\bar{z}_{t-j} \bar{z}_{t-h-k}) &= 0 \text{ except when} \\
 -j &= -h-k \\
 k &= j-h
 \end{aligned}$$

Since $k \geq 0$ it follows that $j \geq h$

$$\begin{aligned}
 \therefore \mathbb{E}(X_t X_{t-h}) &= \sum_{j=h}^{\infty} \psi_j \psi_{j-h} \mathbb{E}(\bar{z}_{t-j}^2) \\
 &= \sigma_z^2 \sum_{j=h}^{\infty} \psi_j \psi_{j-h}.
 \end{aligned}$$

$$6(a) \quad X_t = Z_t + \theta Z_{t-1}$$

$$\begin{aligned} \text{Var}(X_t) &= \text{Var}(Z_t + \theta Z_{t-1}) \\ &= \text{Var}(Z_t) + \theta^2 \text{Var}(Z_{t-1}) \\ &= \sigma^2 (1 + \theta^2) \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_t, X_{t+h}) &= \mathbb{E}[(Z_t + \theta Z_{t-1})(Z_{t+h} + \theta Z_{t+h-1})] \\ &= \theta \mathbb{E}(Z_t^2) \\ &= \sigma^2 \theta \end{aligned}$$

$$\gamma_X(h) = \begin{cases} \sigma^2(1 + \theta^2) & h = 0; \\ \sigma^2 \theta & h = \pm 1; \\ 0 & |h| \geq 2. \end{cases}$$

$$Y_t = W_t + \frac{1}{\theta} W_{t-1} \quad \text{Var}(W_t)_{\text{ind}} = \theta^2 \sigma^2$$

$$\begin{aligned} \text{Var}(Y_t) &= \text{Var}(W_t) + \frac{1}{\theta^2} \text{Var}(W_{t-1}) \\ &= \theta^2 \sigma^2 + \frac{1}{\theta^2} \theta^2 \sigma^2 \\ &= \sigma^2 (1 + \theta^2) \end{aligned}$$

$$\text{Cov}(Y_t, Y_{t+1}) = \sigma^2 \theta$$

We see that $\{X_t\}$ and $\{Y_t\}$ have the same autocovariance function.

$$X_t = Z_t + \theta Z_{t-1}$$

$$Y_t = W_t + \frac{1}{\theta} W_{t-1}$$

$$(i) \quad \theta \rightarrow 0$$

$$X_t = Z_t + \theta Z_{t-1} \rightarrow Z_t$$

We can show this formally using convergence in mean square

$$\begin{aligned} \mathbb{E}((X_t - Z_t)^2) &= \mathbb{E}[(Z_t + \theta Z_{t-1} - Z_t)^2] \\ &= \theta^2 \mathbb{E}(Z_{t-1}^2) \\ &= \theta^2 \sigma_Z^2 \end{aligned}$$

$$\lim_{\theta \rightarrow 0} \mathbb{E}((X_t - Z_t)^2) = 0$$

$\therefore \{X_t\}$ converges to a white noise process $\{Z_t\}$.

Y_t ? $\{W_t\}$ as a WN process with variance $\theta^2 \sigma^2$

$$\text{Define } V_t = \frac{W_t}{\theta} \Rightarrow \text{Var}(V_t) = \frac{1}{\theta^2} \text{Var}(W_t) = \sigma^2$$

$$Y_t = \theta V_t + V_{t-1}$$

$$\begin{aligned} \text{Consider } \mathbb{E}[(Y_t - V_{t-1})^2] &= \mathbb{E}[(\theta V_t + V_{t-1} - V_{t-1})^2] \\ &= \theta^2 \mathbb{E}(V_t^2) \end{aligned}$$

$$= \sigma^2 \theta^2$$

$$\therefore \lim_{\theta \rightarrow 0} \mathbb{E}[(Y_t - V_{t-1})^2] = 0$$

$\Rightarrow$ Convergence in mean square to a white noise process.

$$(ii) \theta = 1$$

$$\left. \begin{aligned} X_t &= Z_t + Z_{t-1} \\ Y_t &= W_t + W_{t-1} \end{aligned} \right\} \begin{array}{l} \text{MA(1) processes.} \\ \underline{\text{Not invertible.}} \end{array}$$

$$8 \quad X_t = 5 + 3t + Z_t - \frac{1}{2} Z_{t-1}$$

$$(a) \mathbb{E}(X_t) = 5 + 3t \quad \leftarrow \text{Non-constant mean}$$

$\therefore \underline{\text{not}} \text{ stationary process.}$

$$\begin{aligned} (b) Y_t &= X_t - X_{t-1} \\ &= \left(5 + 3t + Z_t - \frac{1}{2} Z_{t-1} \right) \\ &\quad - \left(5 + 3(t-1) + Z_{t-1} - \frac{1}{2} Z_{t-2} \right) \\ &= 3 + Z_t - \frac{3}{2} Z_{t-1} + \frac{1}{2} Z_{t-2} \end{aligned}$$

$$\text{MA(2) process.} \quad \mu = 3$$

$$a_1 = -\frac{3}{2}, \quad a_2 = \frac{1}{2}.$$

$$(c) \quad Y_t = \mu + Z_t + a_1 Z_{t-1} + \frac{1}{2} Z_{t-2}$$

$$\begin{aligned} Y_t - \mu &= Z_t + a_1 Z_{t-1} + \frac{1}{2} Z_{t-2} \\ &= \left(1 + a_1 B + \frac{B^2}{2}\right) Z_t \end{aligned}$$

Invertible if $\theta(z) = 0$ has roots outside the unit circle

$$\therefore \text{Set } 1 + a_1 z + \frac{z^2}{2} = 0$$

$$\Rightarrow z = -a_1 \pm \sqrt{a_1^2 - 2}$$

Choose values of a_1 s.t. $|z| > 1$.

Process is invertible if $|a_1| < \frac{3}{2}$.